RETAIL SALES PERFORMANCE & CUSTOMER PURCHASING BEHAVIOUR

An exploratory analysis of transaction-level retail data, 2022–2025

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01 Introduction

Retail businesses make decisions across product assortment, online and in-store operations, payment options and promotions. This study examines a publicly available retail transaction dataset to identify where recorded sales are concentrated, how revenue varies over time and which data-quality issues should be considered before drawing business conclusions. The analysis is descriptive: it identifies patterns in the records rather than proving the causes of customer behaviour.

The dataset contains **12,575 transactions** and **11 original fields**. The reporting period is January 2022 to January 2025. Revenue totals in this report represent the numeric values supplied in the dataset; the currency is not independently verified.

02 Executive findings

TRANSACTIONS	RECORDED SALES	VALID SALES VALUES
12,575	**1,552,071.00**	**11,971 / 12,575**

Category concentration. Butchers contributes the largest recorded category revenue (208,118.00). Differences between categories should be assessed alongside transaction counts and missing sales values.

Channel balance. Online records more revenue (791,401.00) than In-store (760,670.00); this does not necessarily indicate greater profitability.

Payment mix. Cash accounts for the largest recorded payment-method total (537,710.00). The remaining methods also contribute materially.

Data limitations. 604 records lack a transaction value; these are excluded from revenue sums, not treated as confirmed zero-value purchases.

03 Data preparation and analytical approach

The original worksheet was read without changing its source records. Transaction dates were interpreted as dates and Total Spent was converted to numeric values. Missing transaction values were retained as missing; revenue aggregations use only available numeric entries. Category, channel, payment and monthly summaries were calculated independently from the raw worksheet. This avoids assuming that a missing price, quantity or discount flag represents zero or false.

The source workbook also contains prepared analysis sheets. This report uses the Raw_Data sheet for its calculations. Charts were generated from those recalculated summaries; they are not screenshots of an existing dashboard.

04 Findings and visual evidence

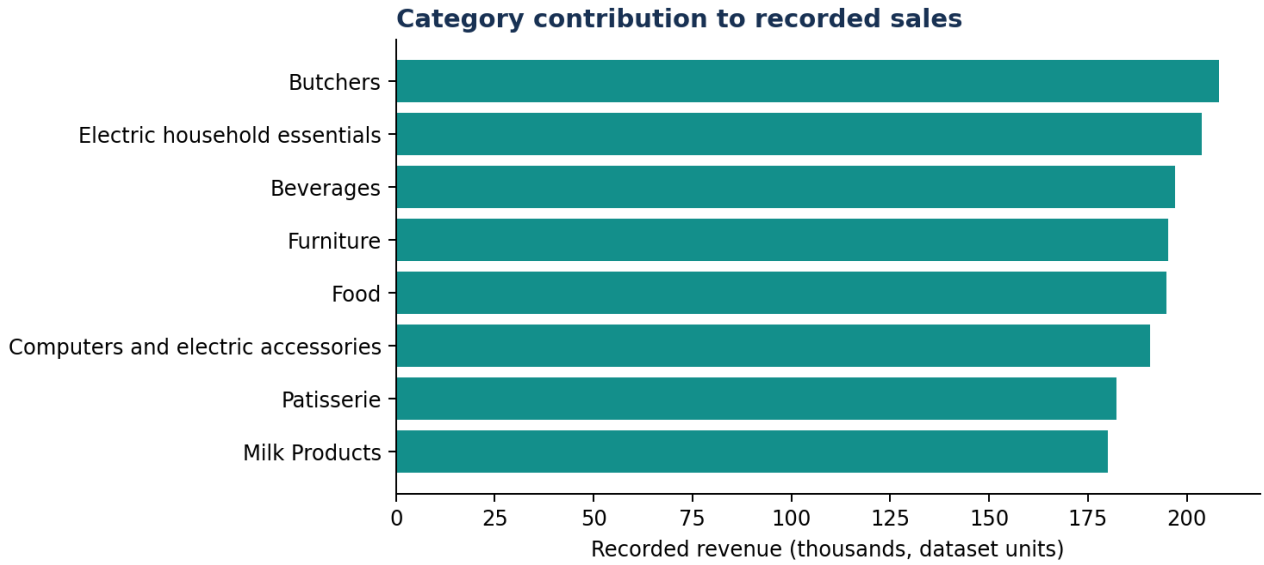


Figure 1. Sum of available Total Spent values by product category. Missing values are excluded.

The category comparison highlights differences in total recorded sales, but the figures should not be read as product profitability. Category-level transaction volumes, margins and missingness would be needed for a stronger commercial interpretation.

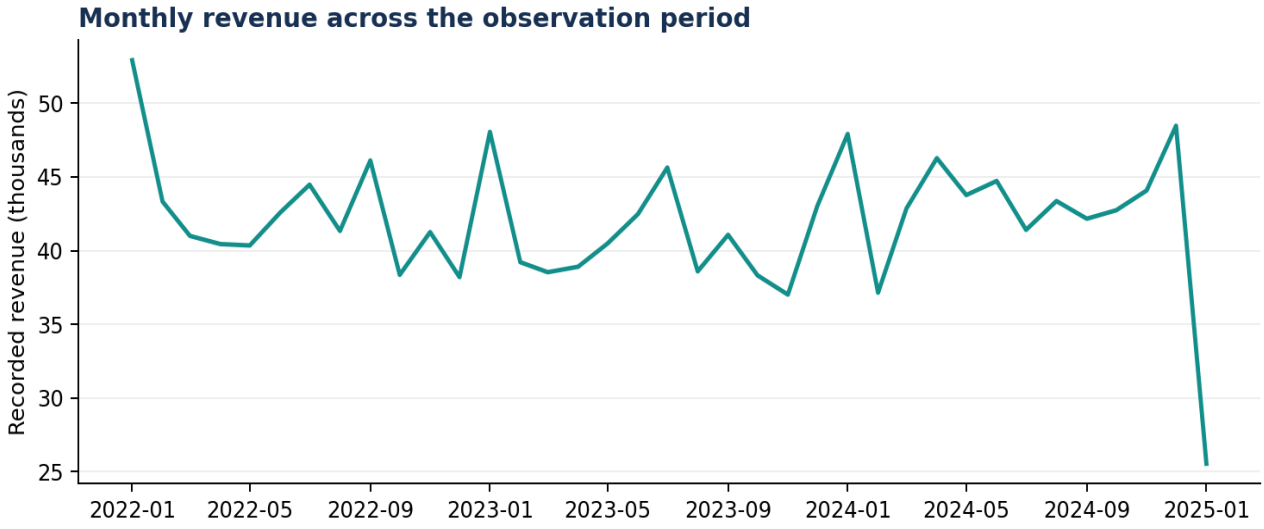


Figure 2. Monthly revenue based on available transaction values. The final month is incomplete and should not be compared directly with full months.

Month-to-month movement is visible across the series. The incomplete final month and any unrecorded sales amounts mean that a decline in the chart is not automatically evidence of reduced underlying demand.

05 Channel and payment behaviour

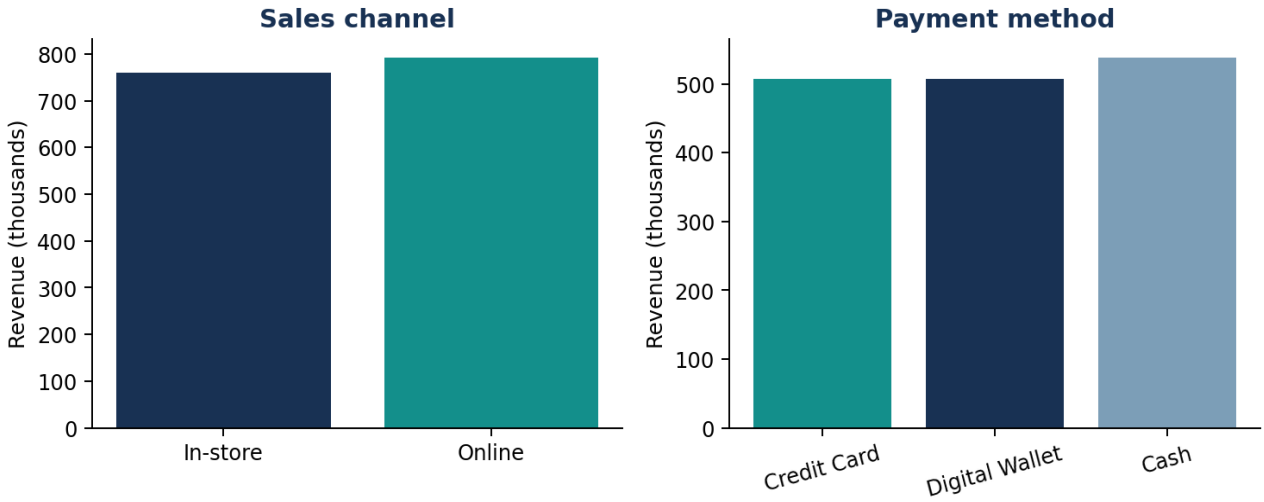


Figure 3. Recorded revenue by transaction location and payment method; the measures represent revenue, not transaction counts.

The two channels both contribute substantially to recorded sales. Payment methods are also relatively diversified. These descriptive comparisons are useful for monitoring operational mix, but the dataset does not establish customer preference or why a particular method was chosen.

06 Data quality and limitations

Field	Missing records	Share of transactions
Discount Applied	4,199	33.4%
Item	1,213	9.6%
Price Per Unit	609	4.8%
Total Spent	604	4.8%
Quantity	604	4.8%

Missing information can affect revenue totals and any comparison between groups. In particular, discount flags should not be imputed as “no discount” without evidence. The dataset also lacks cost and margin information, so this analysis cannot estimate profitability or promotional return on investment.

07 Recommendations

• Create a monthly sales monitoring view that separates complete from partial months and displays missing-value counts alongside revenue.

• Review the product categories with the highest and lowest recorded revenue, using order volume and average transaction value before making assortment decisions.

• Track online and in-store revenue and order counts separately; investigate operational cost before reallocating resources.

• Improve data-entry validation for transaction values and discount indicators before using the dataset for forecasting or promotion evaluation.

08 Conclusion

The analysis identifies meaningful variation across categories, channels, payment methods and months. Its most important qualification is the quality of the underlying records: descriptive findings are useful for prioritising questions, but they do not on their own justify causal claims about discounts or customer behaviour. A next iteration should validate missing values and add order-level and margin measures.

Data source and reproducibility

Ahmedmohamed2003. *Retail Store Sales: Dirty for Data Cleaning*. Kaggle. https://www.kaggle.com/datasets/ahmedmohamed2003/retail-store-sales-dirty-for-data-cleaning

Reproducibility note: figures in this document were recalculated from the Raw_Data worksheet in the supplied retail_store_sales workbook. The document is a drafted analytical report; the author should review and verify its conclusions before academic submission.